Stochastic processes

Corso di "Trasmissione dei segnali digitali"
Laurea Magistrale in Ingegneria Informatica

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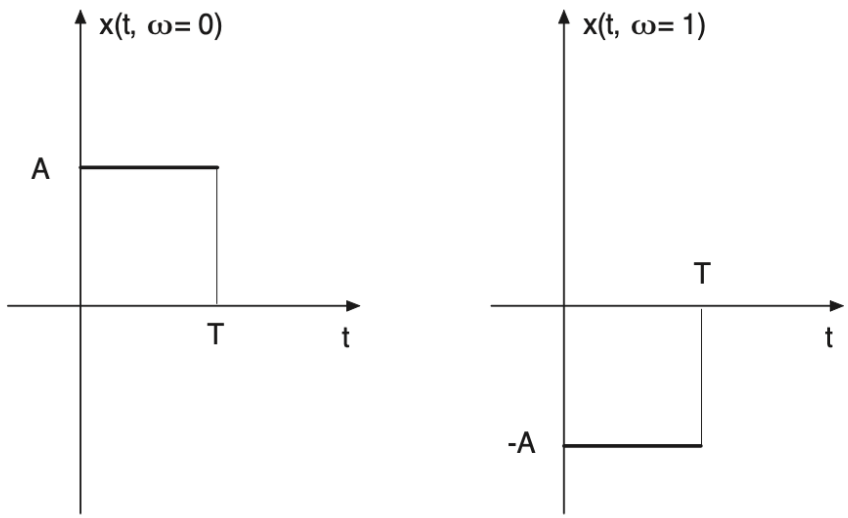
a.a. 2026-27

Introduction

- A *stochastic process* (or *random process* or *random signal*) is a natural extension of the concept of random variable when dealing with signals.
- In many situations the deterministic assumption on time-varying signals is not valid, and it is more appropriate to model signals as *random* functions (for example, thermal noise in electronic circuits or time-varying signals generated by an information source).

Example:
Let $\Omega = \{0, 1\}$, with probabilities $P(\{\omega = 1\}) \stackrel{\triangle}{=} p$ and $P(\{\omega = 0\}) \stackrel{\triangle}{=} q = 1 - p$. We can build a random signal as follows

$$X(t, \omega) = \begin{cases} A\,\text{rect}\left(\dfrac{t - 0.5T}{T}\right), & \text{if } \omega = 0 \\ -A\,\text{rect}\left(\dfrac{t - 0.5T}{T}\right), & \text{if } \omega = 1 \end{cases}$$



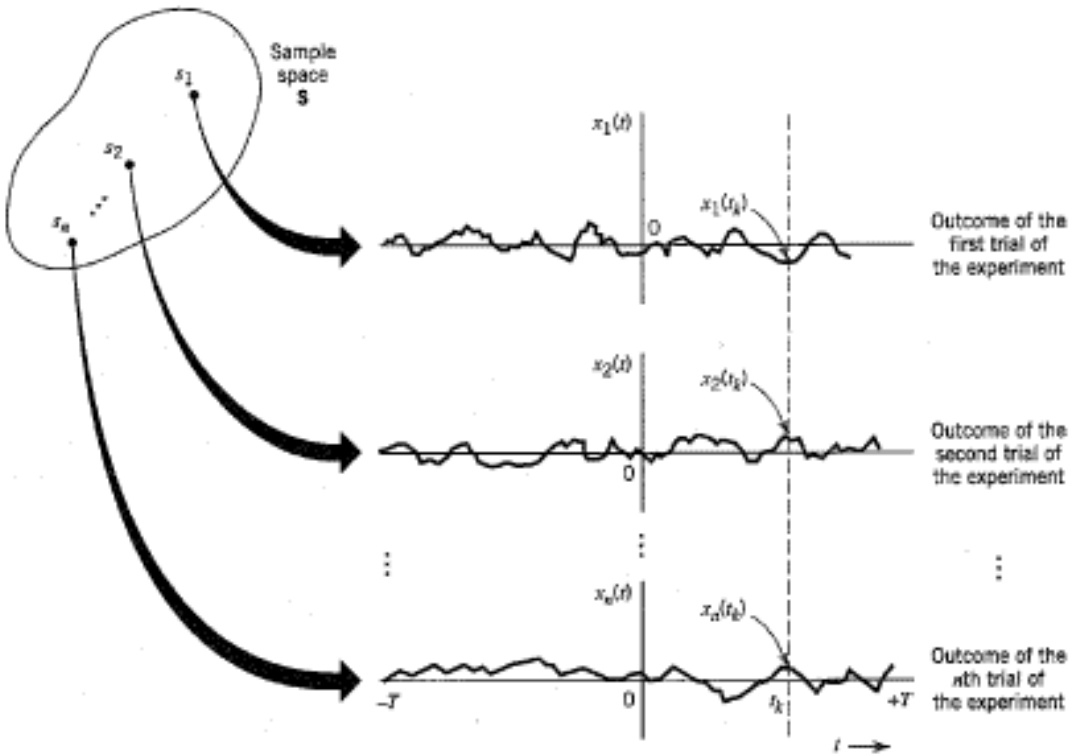
Definition of a stochastic process (1/2)

- A stochastic process $X(t, \omega)$ can be seen as a collection of time signals (also called *realizations* or *sample functions*), corresponding to various outcomes of a random experiment.
  - From this viewpoint, corresponding to each outcome $\omega = \omega_i$ in a probability space $(\Omega, \mathcal{S}, P)$, there exists a deterministic signal $X(t, \omega_i) \equiv x_i(t)$ (this description is similar in spirit to that of random variables).
  - If we fix a time instant t_0, the function $X(t_0, \omega)$ constitutes a random variable.
  - For each time instant t_0 and for each $\omega_i \in \Omega$, $X(t_0, \omega_i)$ represents a number.
- Alternatively, a random process can be viewed as a collection of random variables indexed in time.
  - If the index set is $\mathbb{R}$ the random process is called a *continuous-time* random process.
  - If the index set is $\mathbb{Z}$ the random process is called a *discrete-time* random process (i.e., a sequence of random variables $\{X_i\}_{i=-\infty}^{+\infty}$).
  - This viewpoint is appropriate for mathematical development of the theory of stochastic processes.
- A customary practice is to omit the variable ω in $X(t, \omega)$, for notational simplicity, and use $X(t)$ (or $x(t)$ when no ambiguity arises) to denote a stochastic process.

Definition of a stochastic process (2/2)

Remark

For a random variable, the outcome of a stochastic experiment is mapped into a number. For a stochastic process, the outcome of a stochastic experiment is mapped into a waveform function of time.



Joint distributions of time samples (1/2)

- The statistical description of a stochastic process relies on the cumulative distribution function (CDF) (or, equivalently, probability density function (pdf)) of *vectors* collecting the samples of the random signal.
- Let $X(t_1), X(t_2), \ldots, X(t_k)$ denote the random variables obtained by sampling the process $X(t)$ at times $t_1, t_2, \ldots, t_k$. For $(x_1, x_2, \ldots, x_k) \in \mathbb{R}^k$, the *joint cumulative distribution function (CDF)* of this set of random variables is defined as

$$F_{X(t_1),X(t_2),\ldots,X(t_k)}(x_1, x_2, \ldots, x_k) \stackrel{\triangle}{=} P\left[X(t_1) \leq x_1, X(t_2) \leq x_2, \ldots, X(t_k) \leq x_k\right] \quad (1)$$

  and the *joint probability density function (pdf)* is

$$f_{X(t_1),X(t_2),\ldots,X(t_k)}(x_1, x_2, \ldots, x_k) \stackrel{\triangle}{=} \frac{\partial^n}{\partial x_1 \, \partial x_2 \ldots \partial x_k} F_{X(t_1),X(t_2),\ldots,X(t_k)}(x_1, x_2, \ldots, x_k) \quad (2)$$

- If we know the joint CDF (36) or, equivalently, the joint pdf (2) for any choice of the sampling instant $t_1, t_2, \ldots, t_k$, we say that we know the *statistical characterization of order k* of the stochastic process.

Joint distributions of time samples (2/2)

- A stochastic process is said to be *completely* characterized if the kth-order joint CDF $F_{X(t_1),X(t_2),\ldots,X(t_k)}(x_1, x_2, \ldots, x_k)$ or, equivalently, the kth-order joint pdf $f_{X(t_1),X(t_2),\ldots,X(t_k)}(x_1, x_2, \ldots, x_k)$ is known, for any $k \in \mathbb{N}$ and any choice of sampling instants $t_1, t_2, \ldots, t_k$.
- The collection of CDFs and pdfs must be consistent in the sense that lower-order CDFs and pdfs are obtained as marginals of higher-order CDFs and pdfs, respectively.
- Until now, we have implicitly assumed that the amplitude of the random signal $X(t)$ is *continuous*. If the amplitude of the random signal $X(t)$ is *discrete*, it may be more convenient to resort to the probability distribution function (DF) to characterize the process $X(t)$ (Example: Bernoulli random process).
- Complete characterization of a stochastic process is difficult to achieve in many cases of practical interest $\Rightarrow$ *partial statistical characterization*.

Mean of a stochastic process

- Consider a real-valued stochastic process $X(t)$.
- We define the *mean* of the process $X(t)$ as the expectation of the random variable obtained by sampling the process at t:

$$\mu_X(t) \stackrel{\triangle}{=} \mathrm{E}[X(t)] = \int_{-\infty}^{+\infty} x\, f_{X(t)}(x)\, \mathrm{d}x \tag{3}$$

  where $f_{X(t)}(x)$ is the first-order pdf of $X(t)$.
- In general, $\mu_X(t)$ is a function of time t.

Correlation function of a stochastic process

- We define the *autocorrelation function* (acf) of the stochastic process $X(t)$ as the expectation of the product of two random variables, $X(t_1)$ and $X(t_2)$, obtained by sampling the process $X(t)$ at times t_1 and t_2, as

$$R_{XX}(t_1, t_2) \stackrel{\triangle}{=} \mathrm{E}[X(t_1)\, X(t_2)] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x_1\, x_2\, f_{X(t_1),X(t_2)}(x_1, x_2)\, \mathrm{d}x_1\, \mathrm{d}x_2 \quad (4)$$

  where $f_{X(t_1),X(t_2)}(x_1, x_2)$ is the second-order pdf of $X(t)$.

- In general, $R_{XX}(t_1, t_2)$ is a function of t_1 and t_2 separately.

Covariance function of a stochastic process

- The *autocovariance function* of a process $X(t)$ is defined by

$$\begin{aligned} C_{XX}(t_1, t_2) &\stackrel{\triangle}{=} \mathrm{E}\left\{[X(t_1) - \mu_X(t_1)]\,[X(t_2) - \mu_X(t_2)]\right\} \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} [x_1 - \mu_X(t_1)]\,[x_2 - \mu_X(t_2)]\, f_{X(t_1),X(t_2)}(x_1, x_2)\,\mathrm{d}x_1\,\mathrm{d}x_2 \\ &= R_{XX}(t_1, t_2) - \mu_X(t_1)\,\mu_X(t_2) \end{aligned} \quad (5)$$

- If the mean and the autocorrelation function of $X(t)$ are known, we can determine the autocovariance function.
- In general, $C_{XX}(t_1, t_2)$ is a function of t_1 and t_2 separately.

Stationary stochastic processes (1/3)

- In general, the statistical characterization of a stochastic process $X(t)$ changes by time.
- If the statistical properties of a stochastic process $X(t)$ are not affected, to some extent, by the choice of time origin, the process will be said *stationary* in some sense.
- Different forms of stationarity can be defined.
- Consider a stochastic process $X(t)$ initiated at $t = -\infty$. Let $X(t_1), X(t_2), \ldots, X(t_k)$ denote the random variables obtained by sampling the process $X(t)$ at times $t_1, t_2, \ldots, t_k$. The kth-order joint CDF is given by $F_{X(t_1),X(t_2),\ldots,X(t_k)}(x_1, x_2, \ldots, x_k)$.
- Suppose we shift all the sampling times by a fixed amount τ (*time shift*), obtaining the new set of random variables $X(t_1 + \tau), X(t_2 + \tau), \ldots, X(t_k + \tau)$, whose joint CDF is $F_{X(t_1+\tau),X(t_2+\tau),\ldots,X(t_k+\tau)}(x_1, x_2, \ldots, x_k)$.

Stationary stochastic processes (2/3)

- The stochastic process $X(t)$ is said to be *k-th order stationary* if the following invariance condition

$$F_{X(t_1+\tau),X(t_2+\tau),\ldots,X(t_k+\tau)}(x_1, x_2, \ldots, x_k) \quad = \quad F_{X(t_1),X(t_2),\ldots,X(t_k)}(x_1, x_2, \ldots, x_k) \qquad (6)$$

  holds for all values of time shift τ and any possible choice of $t_1, t_2, \ldots, t_k$.

- A stochastic process $X(t)$ is said *strict-sense* stationary (SSS), or simply strictly stationary, if definition (6) holds for any $k \in \mathbb{N}$ (strongest form of stationarity).

- Often weaker forms of stationarity suffice.

- A stochastic process $X(t)$ is said *first-order stationary* if definition (6) holds for $k = 1$; in this case, letting $\tau = -t$, it results $F_{X(t)}(x) = F_{X(t+\tau)}(x) = F_{X(0)}(x)$, for all $t \in \mathbb{R} \Rightarrow$ the first-order distribution function is *independent* of time t.

- For a first-order stationary process, $\mu_X(t)$ is constant for all time t, i.e.,

$$\mu_X(t) = \int_{-\infty}^{+\infty} x\, f_{X(t)}(x)\, \mathrm{d}x = \int_{-\infty}^{+\infty} x\, f_{X(0)}(x)\, \mathrm{d}x \equiv \mu_X, \quad \forall t \in \mathbb{R}$$

Stationary stochastic processes (3/3)

- A stochastic process $X(t)$ is said *second-order stationary* if definition (6) holds for $k = 2$; in this case, letting $\tau = -t_2$, it results

$$\begin{aligned} F_{X(t_1),X(t_2)}(x_1, x_2) &= F_{X(t_1+\tau),X(t_2+\tau)}(x_1, x_2) \\ &= F_{X(t_1-t_2),X(0)}(x_1, x_2) \end{aligned}$$

  The second-order distribution function of a second-order stationary stochastic process depends only on the *time difference* between the sampling instants and not on the particular times at which the stochastic process is sampled.

- For a second-order stationary process, $f_{X(t_1),X(t_2)}(x_1, x_2)$ depends only on the time difference $t_1 - t_2$ and, thus, $R_{XX}(t_1, t_2)$ also depends only on $t_1 - t_2$, i.e.,

$$R_{XX}(t_1, t_2) \equiv R_{XX}(t_1 - t_2), \quad \forall (t_1, t_2) \in \mathbb{R}^2$$

- It is worth noting that if a stochastic process $X(t)$ is stationary to order k, it is also stationary to order $k - 1$, $k - 2$, and so on (the viceversa is not valid in general).

Wide-sense stationary stochastic processes

- Another important class of stochastic processes is the so-called *wide-sense stationary* (WSS) processes (or *weakly stationary* processes).
- A stochastic process $X(t)$ is said to be WSS if the following two conditions hold:
  - The mean of the process $X(t)$ is constant for all time t.
  - The autocorrelation function of the process $X(t)$ depends solely on the difference between any two times at which the process is sampled.
- Strictly stationarity $\Rightarrow$ wide-sense stationarity.

Autocorrelation function properties (1/2)

- For a WSS process, the autocorrelation function can be expressed as

$$R_{XX}(\tau) \stackrel{\triangle}{=} \mathrm{E}[X(t)\,X(t-\tau)], \quad \forall \tau \in \mathbb{R} \qquad \text{known as } \textit{lag-acf} \tag{7}$$

  where τ, which is called *lag*, denotes the time gap between the process samples in $t_1 = t$ and $t_2 = t_1 - \tau$.

- The lag-acf satisfies the following properties:

  1. The mean-square value of a WSS process $X(t)$ is obtained from $R_{XX}(\tau)$ simply by putting $\tau = 0$ in (7), as shown by $R_{XX}(0) = \mathrm{E}[X^2(t)]$.
  2. $R_{XX}(\tau)$ is an even function of τ, i.e., $R_{XX}(-\tau) = R_{XX}(\tau)$.
  3. The autocorrelation function $R_{XX}(\tau)$ attains its maximum magnitude at $\tau = 0$, i.e., $|R_{XX}(\tau)| \leq R_{XX}(0)$.

- It can be defined a *normalized autocorrelation function* as

$$\rho_{XX}(\tau) \stackrel{\triangle}{=} \frac{R_{XX}(\tau)}{R_{XX}(0)} \tag{8}$$

  whose values are confined to the range $[-1, 1]$, i.e., $|\rho_{XX}(\tau)| \leq 1$.

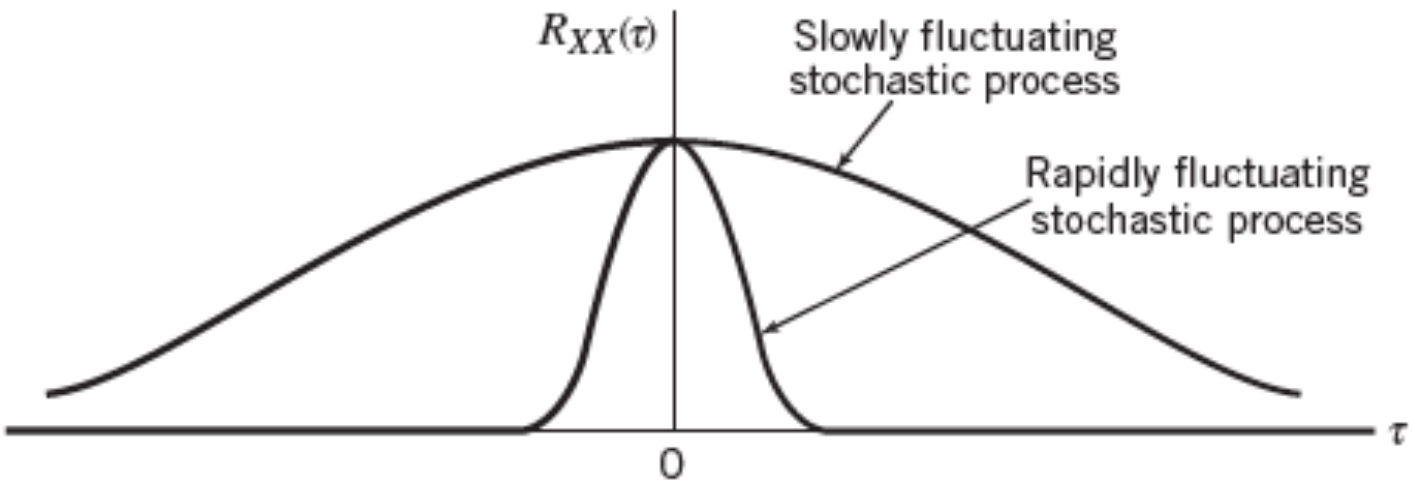
Physical significance of the autocorrelation function (1/2)

- The autocorrelation function $R_{XX}(\tau)$ provides a means of describing the interdependence of two random variables obtained by sampling the stochastic process $X(t)$ at times τ seconds apart.
- *The more rapidly the stochastic process $X(t)$ changes with time, the more rapidly will the autocorrelation function $R_{XX}(\tau)$ decrease from its maximum $R_{XX}(0)$ as τ increases.*
- This behavior of the autocorrelation function may be characterized by a decorrelation time τ_{dec}, such that, for $\tau > \tau_{\text{dec}}$, the magnitude of the autocorrelation function $R_{XX}(\tau)$ remains below some prescribed value.

Physical significance of the autocorrelation function (2/2)

Remark

Illustrating the autocorrelation functions of slowly and rapidly fluctuating stochastic processes.



Crosscorrelation function

- Consider the more general case of two real stochastic processes $X(t)$ and $Y(t)$ with autocorrelation functions $R_{XX}(t,\tau)$ and $R_{YY}(t,\tau)$, respectively. It is possible to define the following lag-*crosscorrelation functions*:

$$R_{XY}(t,\tau) \stackrel{\triangle}{=} \mathrm{E}[X(t)\,Y(t-\tau)] \qquad R_{YX}(t,\tau) \stackrel{\triangle}{=} \mathrm{E}[Y(t)\,X(t-\tau)]\,. \tag{9}$$

- These functions depend on the order of the two processes.
- Two stochastic processes $X(t)$ and $Y(t)$ are said to be *jointly WSS* if each process is WSS and, in addition, $R_{XY}(t,\tau) \equiv R_{XY}(\tau)$.
- In general, the crosscorrelation function is not an even function of the time-shift τ as was true for the autocorrelation function, nor does it have a maximum at the origin, but it does obey the symmetry relationship $R_{XY}(\tau) = R_{YX}(-\tau)$.

Crosscovariance function

- Consider two real stochastic processes $X(t)$ and $Y(t)$ with autocovariance functions $C_{XX}(\tau)$ and $C_{YY}(\tau)$, respectively. It is possible to define the following lag-*crosscovariance function*:

$$C_{XY}(t,\tau) \stackrel{\triangle}{=} \mathrm{E}\left\{[X(t)-\mu_X(t)]\,[Y(t-\tau)-\mu_Y(t-\tau)]\right\} \tag{10}$$

$$C_{YX}(t,\tau) \stackrel{\triangle}{=} \mathrm{E}\left\{[Y(t)-\mu_Y(t)]\,[X(t-\tau)-\mu_X(t-\tau)]\right\} \tag{11}$$

- These functions depend on the order of the two processes.
- If $X(t)$ and $Y(t)$ are zero-mean stochastic processes, i.e., $\mu_X(t) = \mu_Y(t) = 0$, $\forall t \in \mathbb{R}$, then $C_{XY}(t,\tau) = R_{XY}(t,\tau)$ (and $C_{YX}(t,\tau) = R_{YX}(t,\tau)$).
- If the stochastic processes $X(t)$ and $Y(t)$ are jointly WSS then the crosscovariance function becomes $C_{XY}(t,\tau) = C_{XY}(\tau)$.

Independent, incoherent, and uncorrelated stochastic processes

- Two stochastic processes $X(t)$ and $Y(t)$ are *independent random processes* if the two sets of random variables $\mathcal{X} = \{X(t_1), X(t_2), \ldots, X(t_k)\}$ and $\mathcal{Y} = \{Y(t_1^{'}), Y(t_2^{'}), \ldots, Y(t_j^{'})\}$ are independent for all $k, j \in \mathbb{N}$ and all choices of $t_1, t_2, \ldots, t_k$ and $t_1^{'}, t_2^{'}, \ldots, t_j^{'}$.
- If $R_{XY}(\tau) = 0$ for any $\tau \in \mathbb{R}$, the two stochastic processes $X(t)$ and $Y(t)$ are said *incoherent*.
- If $C_{XY}(\tau) = 0$ for any $\tau \in \mathbb{R}$, the two stochastic processes $X(t)$ and $Y(t)$ are said *uncorrelated*.
- It can be proven that if $X(t)$ and $Y(t)$ are independent, then they are uncorrelated.
- If $\mu_X = \mu_Y = 0$, then incoherence is equivalent to uncorrelatedness.

Complex stochastic processes

- A complex stochastic process can be represented as

$$Z(t) = X(t) + j\,Y(t)$$

- The complete statistical characterization of $Z(t)$ may be obtained through the *joint* statistical characterization of its real $X(t)$ and imaginary $Y(t)$ part.

- Regarding the partial statistical characterization of $Z(t)$ (i.e., mean and acf), it results

$$\mu_Z(t) \stackrel{\triangle}{=} \mathrm{E}[Z(t)] = \mathrm{E}[X(t) + j\,Y(t)] = \mu_X(t) + j\,\mu_Y(t) \tag{12}$$

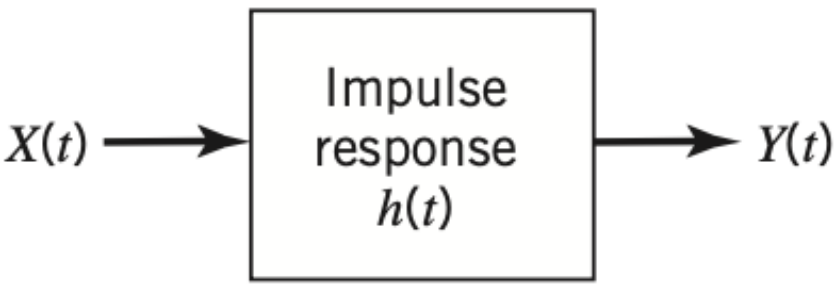
$$R_{ZZ}(t_1, t_2) \stackrel{\triangle}{=} \mathrm{E}[Z(t_1)\,Z^*(t_2)] \quad \text{or} \quad R_{ZZ}(t, \tau) \stackrel{\triangle}{=} \mathrm{E}[Z(t)\,Z^*(t-\tau)] \tag{13}$$

  with $Z^*(\cdot)$ denoting the complex conjugate of $Z(\cdot)$.

- To be rigorous we should introduce another second-order moment, the so-called *conjugate autocorrelation function* $R_{ZZ^*}(t_1, t_2) \stackrel{\triangle}{=} \mathrm{E}[Z(t_1)\,Z(t_2)]$, which however is null for most processes.

Transmission of a WSS Process through a LTI Filter

- Let $X(t)$ denote a stochastic process that is applied as input to a linear time-invariant (LTI) deterministic filter, with impulse response $h(t)$, and let $Y(t)$ denote the corresponding stochastic output.



- The process at the output of the LTI filter can be written through the *convolution integral*:

$$Y(t) = \int_{-\infty}^{+\infty} h(\tau_1)\, X(t-\tau_1)\, \mathrm{d}\tau_1$$

- In general, it is difficult to determine the probability distribution of $Y(t)$, even when the probability distribution of the input $X(t)$ is known.

- In many cases, it suffices to calculate the mean and autocorrelation functions of $Y(t)$ in terms of those of $X(t)$, assuming that $X(t)$ is WSS.

Mean of the system output (1/2)

- The mean of $Y(t)$ is

$$\mu_Y(t) = \mathrm{E}[Y(t)] = \mathrm{E}\left[\int_{-\infty}^{+\infty} h(\tau_1)\, X(t-\tau_1)\, \mathrm{d}\tau_1\right] \tag{14}$$

- Provided that $\mathrm{E}[X(t)]$ is finite for all t and the filter is stable, we may interchange the order of expectation and integration:

$$\begin{aligned}\mu_Y(t) &= \int_{-\infty}^{+\infty} h(\tau_1)\, \mathrm{E}\left[X(t-\tau_1)\right]\, \mathrm{d}\tau_1 \\ &= \int_{-\infty}^{+\infty} h(\tau_1)\, \mu_X(t-\tau_1)\, \mathrm{d}\tau_1 = \mu_X(t) * h(t)\end{aligned} \tag{15}$$

  - $\mu_Y(t)$ is the convolution between the mean of the input stochastic process and the impulse response of the filter.

Mean of the system output (2/2)

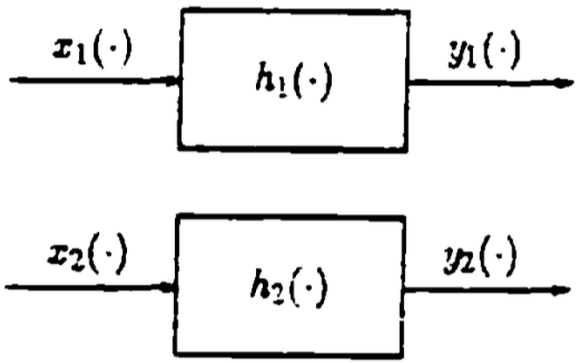
- When the input stochastic process $X(t)$ is weakly stationary, i.e., $\mu_X(t) \equiv \mu_X$, one has

$$\mu_Y(t) \equiv \mu_Y = \mu_X \int_{-\infty}^{+\infty} h(\tau_1) \mathrm{d}\tau_1 = \mu_X \, H(0) \tag{16}$$

  - $H(0)$ is the zero-frequency response of the system.

- *The mean of the stochastic process $Y(t)$ produced at the output of a linear time-invariant filter in response to a WSS process $X(t)$ is equal to the mean of $X(t)$ multiplied by the zero-frequency response of the filter.*

Correlation functions of output response of LTI system (1/3)



Let us compute the crosscorrelation function of the two output responses $Y_1(t)$ and $Y_2(t)$ of an LTI system, i.e., $R_{Y_1 Y_2}(t_1, t_2) = \mathrm{E}[Y_1(t_1)\, Y_2(t_2)]$.

- We may apply the input-output relation for an LTI system and, interchanging the order of integration (assuming it possible), we obtain

$$R_{Y_1 Y_2}(t_1, t_2) = \mathrm{E}\left[\int_{-\infty}^{+\infty} h_1(\tau_1)\, X_1(t_1 - \tau_1)\, \mathrm{d}\tau_1 \int_{-\infty}^{+\infty} h_2(\tau_2)\, X_2(t_2 - \tau_2)\, \mathrm{d}\tau_2\right] \tag{17}$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_1(\tau_1)\, h_2(\tau_2)\, \mathrm{E}[X_1(t_1 - \tau_1)\, X_2(t_2 - \tau_2)]\, \mathrm{d}\tau_1\, \mathrm{d}\tau_2$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_1(\tau_1)\, h_2(\tau_2)\, R_{X_1 X_2}(t_1 - \tau_1, t_2 - \tau_2)\, \mathrm{d}\tau_1\, \mathrm{d}\tau_2 \tag{18}$$

Correlation functions of output response of LTI system (2/3)

- When $X_1(t)$ and $X_2(t)$ are jointly WSS processes, the crosscorrelation function is only a function of the difference between $t_1 - \tau_1$ and $t_2 - \tau_2$. Thus, putting $\tau = t_1 - t_2$ in (18), one has

$$\begin{aligned} R_{Y_1Y_2}(t_1, t_2) &\equiv R_{Y_1Y_2}(\tau) \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_1(\tau_1)\, h_2(\tau_2)\, R_{X_1X_2}(\tau - \tau_1 + \tau_2)\, \mathrm{d}\tau_1\, \mathrm{d}\tau_2 \qquad (19) \\ &= R_{X_1X_2}(\tau) * h_1(\tau) * h_2(-\tau) = R_{X_1X_2}(\tau) * r_{h_1h_2}(\tau) \qquad (20) \end{aligned}$$

  which solely depends on the lag τ.

- It is worth noting that if the two inputs are incoherent, then also the outputs will be, i.e., an LTI processing saves incoherence.

Correlation functions of output response of LTI system (3/3)

- Particularizing the relation (18) to the case $h_1(t) = h_2(t) = h(t)$ and $X_1(t) = X_2(t) = X(t)$, one easily obtains the input-output relation for autocorrelation functions, i.e.,

$$R_{YY}(t_1, t_2) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h(\tau_1)\, h(\tau_2)\, R_{XX}(t_1 - \tau_1, t_2 - \tau_2)\, \mathrm{d}\tau_1\, \mathrm{d}\tau_2$$

- If the input process $X(t)$ is WSS, then the output autocorrelation function can be expressed as

$$R_{YY}(\tau) = R_{XX}(\tau) * h(\tau) * h(-\tau) = R_{XX}(\tau) * r_{hh}(\tau)$$

- If the input of a stable LTI system is WSS, the output of the system is also WSS.

- Equations (18) and (20) allow one to derive also the crosscorrelation functions $R_{YX}(\cdot)$ and $R_{XY}(\cdot)$, as particular cases. By letting, indeed, in (20) $Y_1(t) = Y(t)$, $Y_2(t) = X(t)$, $h_1(t) = h(t)$, $h_2(t) = \delta(t)$, one has $R_{YX}(\tau) = R_{XX}(\tau) * h(\tau)$. Similarly, one can obtain $R_{XY}(\tau) = R_{XX}(\tau) * h(-\tau)$.

Power Spectral Density (PSD)

- The power spectral density (PSD) of a stochastic process $X(t)$ is defined as

$$S_{XX}(f) \stackrel{\triangle}{=} \lim_{T \to +\infty} \frac{1}{2\,T} \mathrm{E}[|X_T(f)|^2]$$

  where $X_T(f)$ represents the Fourier transform of the *time-windowed* version of $X(t)$, i.e., $X_T(t) = X(t)\,\mathrm{rect}[t/(2T)]$.

- It can be proven that the PSD $S_{XX}(f)$ of a WSS random process $X(t)$ is the Fourier transform of its autocorrelation function $R_{XX}(\tau)$. This result is known as *Wiener-Khinchin relation*.

- The mean-square value of the random process $X(t)$ can be evaluated as

$$\mathrm{E}[X^2(t)] = R_{XX}(0) = \int_{-\infty}^{+\infty} S_{XX}(f)\,\mathrm{d}f = P_{\mathrm{X}} \tag{21}$$

  with P_{X} denoting the power of the stochastic WSS process $X(t)$.

Power spectral density and LTI systems

- Let $S_{YY}(f)$ denote the power spectral density of the output stochastic process $Y(t)$ obtained by passing the WSS process $X(t)$ through a linear time-invariant filter of frequency response $H(f)$.
- By recalling that the PSD of a WSS process is equal to the Fourier transform of its autocorrelation function (Wiener-Khinchin relation) and that $R_{YY}(\tau) = R_{XX}(\tau) * h(\tau) * h(-\tau)$, we obtain

$$S_{YY}(f) = S_{XX}(f)\,|H(f)|^2 \tag{22}$$

  where $H(f)$ is the frequency response of the filter.
- The power spectral density of the output process $Y(t)$ equals the power spectral density of the input process $X(t)$, multiplied by the squared magnitude response of the filter.

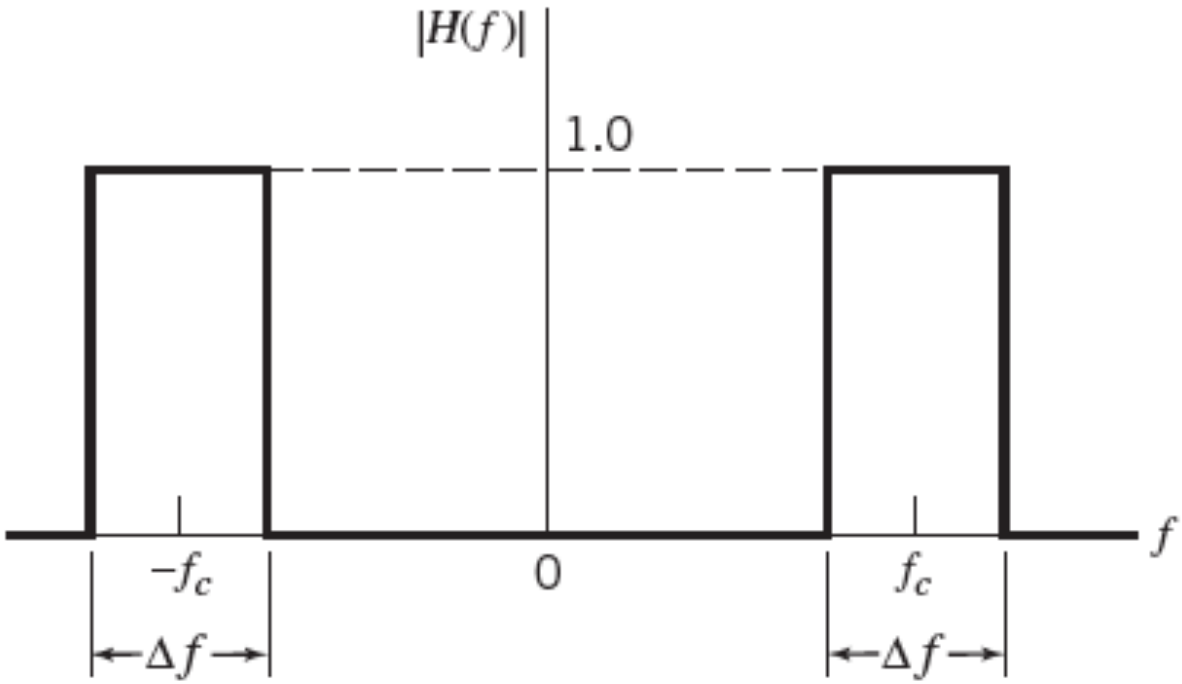
Physical Significance of the PSD (1/2)

- Suppose that the WSS process $X(t)$ is passed through an ideal narrowband filter with a magnitude response $|H(f)|$ centered about the frequency f_c, as in the figure in the next slide, where Δf is the bandwidth of the filter.
- From (22), we find that if the bandwidth Δf is made sufficiently small compared with f_c and $S_{XX}(f)$ is a continuous function of the frequency f, then the power ΔP_X of the spectral components of $X(t)$ belonging to the frequency interval of length $2\Delta f$ is approximately given by $\Delta P_X \approx (2\,\Delta f)\, S_{XX}(f)$, with $f_c \equiv f$ for generality.
  - $S_{XX}(f)$ is measured in Watts per hertz (W/Hz).

Physical Significance of the PSD (2/2)

Remark

Magnitude response of ideal narrowband filter.



Properties of the PSD (1/2)

- *Property 1*: The zero-frequency value of the power spectral density of a WSS process equals the total area under the graph of the autocorrelation function, that is,

$$S_{XX}(0) = \int_{-\infty}^{+\infty} R_{XX}(\tau)\,\mathrm{d}\tau \tag{23}$$

- *Property 2*: The mean-square value of a WSS process $X(t)$ equals the total area under the graph of the power spectral density of the process, that is,

$$\mathrm{E}[X^2(t)] = \int_{-\infty}^{+\infty} S_{XX}(f)\,\mathrm{d}f = R_{XX}(0) \tag{24}$$

- *Property 3*: The power spectral density of a stationary process $X(t)$ is nonnegative, that is, $S_{XX}(f) \geq 0$, $\forall f \in \mathbb{R}$.

Properties of the PSD (2/2)

- *Property 4*: The power spectral density of a real-valued weakly stationary process is an even function, i.e., $S_{XX}(-f) = S_{XX}(f)$.
- *Property 5*: The power spectral density, appropriately normalized, has the properties associated with a pdf:

$$p_{XX}(f) \stackrel{\triangle}{=} \frac{S_{XX}(f)}{\int_{-\infty}^{+\infty} S_{XX}(f)\,\mathrm{d}f} \tag{25}$$

  - $p_{XX}(f) \geq 0$, $\forall f \in \mathbb{R}$, and the total area under $p_{XX}(f)$ is unity.
- **Example** - Compute power spectral density of a linearly modulated signal.

Cross-spectral Densities

- *Cross-spectral density* characterizes the frequency-domain relationships between two random processes.
- The cross-spectral density $S_{XY}(f)$ of two processes $X(t)$ and $Y(t)$ is defined as

$$S_{XY}(f) \triangleq \lim_{T\to+\infty} \frac{1}{2T}\mathrm{E}[X_T(f)Y_T^*(f)]$$

  where $X_T(f)$ and $Y_T(f)$ are the Fourier transforms of the *time-windowed* versions of $X(t)$ and $Y(t)$, respectively.
- For two jointly WSS processes, the cross-spectral density is the Fourier transform of their cross-correlation function:

$$S_{XY}(f) = \int_{-\infty}^{+\infty} R_{XY}(\tau)e^{-j2\pi f\tau}\,\mathrm{d}\tau. \tag{26}$$

- Unlike the power spectral density, the cross-spectral density is not necessarily real or nonnegative.
- For complex-valued processes, the relation $R_{XY}(\tau) = R_{YX}(-\tau)$ generalizes to $R_{XY}(\tau) = R_{YX}^*(-\tau)$. Consequently, the cross-spectral densities satisfy

$$S_{XY}(f) = S_{YX}^*(f).$$

Sum of Two Weakly Stationary Processes (1/2)

- Suppose that the stochastic processes $X(t)$ and $Y(t)$ have zero mean and let their sum be denoted by $Z(t) = X(t) + Y(t)$. The problem is to determine the power spectral density of the process $Z(t)$.
- The autocorrelation function of $Z(t)$ is given by

$$\begin{aligned} R_{ZZ}(t_1, t_2) &= \mathrm{E}[Z(t_1)\, Z(t_2)] = \mathrm{E}\left\{[X(t_1) + Y(t_1)]\,[X(t_2) + Y(t_2)]\right\} \\ &= R_{XX}(t_1, t_2) + R_{XY}(t_1, t_2) + R_{YX}(t_1, t_2) + R_{YY}(t_1, t_2) \end{aligned} \tag{27}$$

- Defining $\tau = t_1 - t_2$ and assuming the joint weakly stationarity of the two processes, we may go on to write

$$R_{ZZ}(\tau) = R_{XX}(\tau) + R_{XY}(\tau) + R_{YX}(\tau) + R_{YY}(\tau) \tag{28}$$

Sum of Two Weakly Stationary Processes (2/2)

- Taking the Fourier transform of both sides of (28), we get

$$S_{ZZ}(f) = S_{XX}(f) + S_{XY}(f) + S_{YX}(f) + S_{YY}(f) \tag{29}$$

- When the WSS processes $X(t)$ and $Y(t)$ are incoherent, i.e., $R_{XY}(\tau) = R_{YX}(\tau) = 0$, $\forall \tau \in \mathbb{R}$, the cross-spectral densities $S_{XY}(f)$ and $S_{YX}(f)$ are zero, in which case (29) reduces to

$$S_{ZZ}(f) = S_{XX}(f) + S_{YY}(f) \tag{30}$$

- *The PSD of the sum of incoherent WSS processes (or, equivalently, zero-mean uncorrelated processes) is equal to the sum of their individual PSDs.*

The Gaussian Process (1/3)

- The *Gaussian process* is the most frequently encountered random process in the study of communication systems thanks to its practical applicability and mathematical tractability.
- Consider the random variables $X(t_1), X(t_2), \ldots, X(t_n)$, obtained by sampling a stochastic process $X(t)$ at times $t_1, t_2, \ldots, t_n$. If the process $X(t)$ is Gaussian, then this set of random variables is jointly Gaussian for any n, with their nth-order joint pdf being completely determined by specifying the set of means

$$\mu_{X(t_i)} \stackrel{\triangle}{=} \mathrm{E}[X(t_i)]\,, \quad \text{for } i \in \{1, 2, \ldots, n\} \tag{31}$$

and, for $k, i \in \{1, 2, \ldots, n\}$, the set of covariance functions

$$C_{XX}(t_k, t_i) \stackrel{\triangle}{=} \mathrm{E}\left\{[X(t_k) - \mu_{X(t_k)}]\,[X(t_i) - \mu_{X(t_i)}]\right\} \tag{32}$$

- Let $\mathbf{X} \stackrel{\triangle}{=} [X(t_1), X(t_2), \ldots, X(t_n)]^{\mathsf{T}}$ be the random vector collecting all the random variables and denote with $\mathbf{x} \stackrel{\triangle}{=} [x_1, x_2, \ldots, x_n]^{\mathsf{T}}$ its sample value (the superscript T denotes transposition).

The Gaussian Process (2/3)

- The random vector $\mathbf{X}$ has a multivariate Gaussian distribution, defined in matrix form as

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n \det(\boldsymbol{\Sigma})}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu})} \tag{33}$$

  where $\boldsymbol{\mu} \stackrel{\triangle}{=} [\mu_{X(t_1)}, \mu_{X(t_2)}, \ldots, \mu_{X(t_n)}]^{\mathsf{T}}$ is the mean vector, $\boldsymbol{\Sigma}$ is the covariance matrix whose (k, i) element is $\{\boldsymbol{\Sigma}\}_{k,i} = C_{XX}(t_k, t_i)$, for $k, i \in \{1, 2, \ldots, n\}$, and $\boldsymbol{\Sigma}^{-1}$ is the matrix inverse of $\boldsymbol{\Sigma}$.

- The covariance matrix $\boldsymbol{\Sigma}$ is symmetric (i.e., $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^{\mathsf{T}}$) and nonnegative definite (i.e., $\mathbf{u}^{\mathsf{T}} \boldsymbol{\Sigma} \mathbf{u} \geq 0$, $\forall \mathbf{u} \in \mathbb{R}^n$). For a nondegenerate Gaussian process, $\boldsymbol{\Sigma}$ is positive definite (i.e., $\mathbf{u}^{\mathsf{T}} \boldsymbol{\Sigma} \mathbf{u} > 0$, $\forall \mathbf{u} \in \mathbb{R}^n - \{\mathbf{0}\}$), in which case the covariance matrix is invertible.

The Gaussian Process (3/3)

- The stochastic processes produced by physical phenomena are often such that a Gaussian model is appropriate. This model to describe physical phenomena is confirmed by experiments.
- Last, but by no means least, the central limit theorem provides mathematical justification for the Gaussian distribution.
- The frequent occurrence of physical phenomena for which a Gaussian model is appropriate and the ease with which a Gaussian process is handled mathematically make the Gaussian process very important in the study of communication systems.

Properties of Gaussian Processes

- If a Gaussian process $X(t)$ is applied to a stable linear filter, then the stochastic process $Y(t)$ developed at the output of the filter is also Gaussian.
- If a Gaussian process is WSS (also known as weakly stationary), then the process is also strictly stationary.
- If the random variables $X(t_1), X(t_2), \ldots, X(t_n)$, obtained by sampling a stochastic process $X(t)$ at times $t_1, t_2, \ldots, t_n$, are uncorrelated, i.e., $C_{XX}(t_k, t_i) = 0$, for $k \neq i$, then these random variables are statistically independent.
  - The uncorrelatedness $X(t_1), X(t_2), \ldots, X(t_n)$ means that the covariance matrix is reduced to a diagonal matrix, i.e., $\mathbf{\Sigma} = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2)$, where σ_i^2 is the variance of $X(t_i) \Rightarrow$ the pdf in (33) simplifies to

$$f_{\mathbf{X}}(\mathbf{x}) = \prod_{i=1}^{n} f_{X(t_i)}(x_i) \tag{34}$$

  - The joint pdf of the set of random variables is expressed as the product of the probability density functions of the individual random variables in the set.

Discrete-time random processes (1/2)

- So far, we have considered *continuous-time* random processes, i.e., $X(t;\omega)$ with $t \in \mathbb{R}$.
- If the time variable belongs to $\mathbb{Z}$, the random process is called a *discrete-time* random process and is denoted by $X(n;\omega)$.
- A discrete-time random process is therefore a sequence of random variables $\{X(n)\}_{n\in\mathbb{Z}}$ (where the dependence on ω is omitted). For $n \in \{0, 1, \ldots, N-1\}$, it can be represented by a *random vector*.
- CDFs, PDFs, and distribution functions for discrete-time random processes can be defined as in the continuous-time case by replacing t with n.
- The mean and cross-correlation functions are defined as

$$\mu_X(n) \triangleq \mathrm{E}[X(n)] \tag{35}$$

$$R_{XY}(n,k) \triangleq \mathrm{E}[X(n)Y^*(k)]. \tag{36}$$

Discrete-time random processes (2/2)

- The cross-covariance function of two discrete-time random processes is defined as

$$C_{XY}(n,k) \stackrel{\triangle}{=} \mathrm{E}\{[X(n)-\mu_X(n)][Y(k)-\mu_Y(k)]^*\} = R_{XY}(n,k) - \mu_X(n)\mu_Y^*(k). \quad (37)$$

- By letting $X(n) \equiv Y(n)$ in (36) and (37), we obtain the autocorrelation and autocovariance functions.
- For two jointly WSS discrete-time random processes, the cross-spectral density is

$$S_{XY}(\nu) \stackrel{\triangle}{=} \lim_{N\to+\infty} \frac{1}{2N+1}\mathrm{E}[X_N(\nu)Y_N^*(\nu)]$$

where, for $Z \in \{X, Y\}$, $Z_N(\nu) \stackrel{\triangle}{=} \sum_{n=-N}^{N} Z(n)e^{-j2\pi\nu n}$ is the discrete-time Fourier transform of the *truncated process* $Z(n)$.

- By letting $X(n) \equiv Y(n)$, we obtain the PSD

$$S_{XX}(\nu) \stackrel{\triangle}{=} \lim_{N\to+\infty} \frac{1}{2N+1}\mathrm{E}[|X_N(\nu)|^2].$$

- By the Wiener-Khinchin theorem, the PSD of a discrete-time WSS process satisfies

$$S_{XX}(\nu) = \sum_{m=-\infty}^{+\infty} R_{XX}(m)e^{-j2\pi\nu m}.$$